PREP 4, LESSONS 7–8

Specific heat capacity

$\Delta E = m\,c\,\Delta\theta$; measuring specific heat capacity (required practical)

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define internal energy. DN 190 [1]

The total energy stored by the particles that make up a system.

2 Define specific latent heat. DN 202 [1]

The energy needed to change the state of 1 kg of a substance with no change in temperature.

3 Describe how to read a measuring cylinder accurately. DN 698 [1]

Read the bottom of the meniscus at eye level.

Key ideas LESSONS 7–8

Show your working

4 2.0 kg of water is heated from 20 °C to 50 °C. Specific heat capacity of water = 4200 J/kg °C. Use $\Delta E = m\,c\,\Delta\theta$
Calculate the energy transferred to the water. [3]

$\Delta\theta = 50 - 20 = 30\,°C$ ✓

$\Delta E = 2.0 \times 4200 \times 30$ ✓

$\Delta E = 252\,000\,J$ ✓

5 The unit of specific heat capacity is J/kg | **J/kg °C** | J/°C [1]

6 A student heats a metal block with an immersion heater to find its specific heat capacity.

a Give the reason why the block is wrapped in insulation. [1]

To reduce energy transfer to the surroundings ✓

b The student's value is higher than the true value. Explain why. [2]

Some energy is transferred to the surroundings ✓

so the temperature rise is smaller than it should be ✓